Order Management System Using SQL

Project Documentation

1. Project Overview

This project manages customers, products, orders, and order details using MySQL. It demonstrates table creation, relationships, constraints, CRUD operations, and sales reports.

2. Table Design

Customer

Customer_ID (PK), Customer_Name (NOT NULL), Email (UNIQUE, NOT NULL), Phone (UNIQUE), City (NOT NULL).

Product

Product_ID (PK), Product_Name (UNIQUE, NOT NULL), Price (CHECK Price >= 0), Stock_Quantity (CHECK Stock_Quantity >= 0).

Orders

Order_ID (PK), Customer_ID (FK), Order_Date (NOT NULL), Total_Amount (CHECK Total_Amount >= 0), Order_Status (CHECK: Pending, Shipped, Delivered, Cancelled).

Order_Details

Order_Detail_ID (PK), Order_ID (FK), Product_ID (FK), Quantity (CHECK Quantity > 0), Price (CHECK Price >= 0), UNIQUE (Order_ID, Product_ID).

3. ER Diagram

CUSTOMER (1) ————< ORDERS (1) ————< ORDER_DETAILS >———— (1) PRODUCT

Relationship meaning:
• One customer can place many orders.
• One order can contain multiple order-detail rows.
• One product can appear in many order-detail rows.
• Order_Details connects Orders and Product.

Text ER diagram for reference:
[Customer: Customer_ID PK] 1 ---- many [Orders: Order_ID PK, Customer_ID FK]
[Orders: Order_ID PK] 1 ---- many [Order_Details: Order_Detail_ID PK, Order_ID FK, Product_ID FK]
[Product: Product_ID PK] 1 ---- many [Order_Details]

4. Constraints Used

PRIMARY KEY uniquely identifies each row.

FOREIGN KEY maintains valid relationships between tables.

NOT NULL makes required fields mandatory.

UNIQUE prevents duplicate email addresses, product names, and duplicate products within the same order.

CHECK validates values such as non-negative prices and valid order statuses.

ON DELETE CASCADE on Order_Details removes detail rows when an order is deleted.

5. Sample Data

The SQL script inserts four customers, five products, five initial orders, and eight order-detail rows. It then recalculates order totals and creates one additional order for demonstration.

Run the INSERT statements in the SQL script to view the exact sample data.

6. Order Management Operations

Create an order by inserting a row into Orders.

Add multiple products by inserting multiple rows into Order_Details using the same Order_ID.

Calculate Total_Amount using SUM(Quantity × Price).

Update status to Pending, Shipped, Delivered, or Cancelled.

Update an order-detail quantity and recalculate the order total.

Delete a cancelled order; its Order_Details rows are removed through ON DELETE CASCADE.

Use JOIN queries to display customer, order, product, quantity, price, and line total.

7. Customer Order History and Purchase Analysis

Customer order history: Lists each customer and their orders, including customers who have no orders.

Product-wise order report: Shows the number of orders, quantity sold, and sales amount for each product.

Times each product was ordered: Counts distinct orders that include each product.

Total quantity sold: Adds quantities from non-cancelled orders.

Customers with maximum orders: Returns all customers tied for the highest order count.

Total spending by customer: Adds non-cancelled order totals for each customer.

Average order value: Calculates the average total amount for non-cancelled orders.

Total sales amount: Adds totals for all non-cancelled orders.

8. Query Output Screenshots

After running the SQL script in MySQL Workbench, take screenshots of the result grid for these queries:

SELECT * FROM Customer;

SELECT * FROM Product;

SELECT * FROM Orders;

SELECT * FROM Order_Details;

The complete order information JOIN query.

Customer order history query.

Product-wise order report.

Total sales amount and average order value queries.

Insert your real MySQL Workbench screenshots here before submitting. Do not use this documentation text as a substitute for actual query-output screenshots.

9. Conclusion

The Order Management System demonstrates relational database design and SQL operations. Primary and foreign keys maintain data integrity, constraints validate data, and report queries summarize customer purchases and product sales.